SGL-omni

Redesigning Inference Framework for Multi-Stage Generative Models

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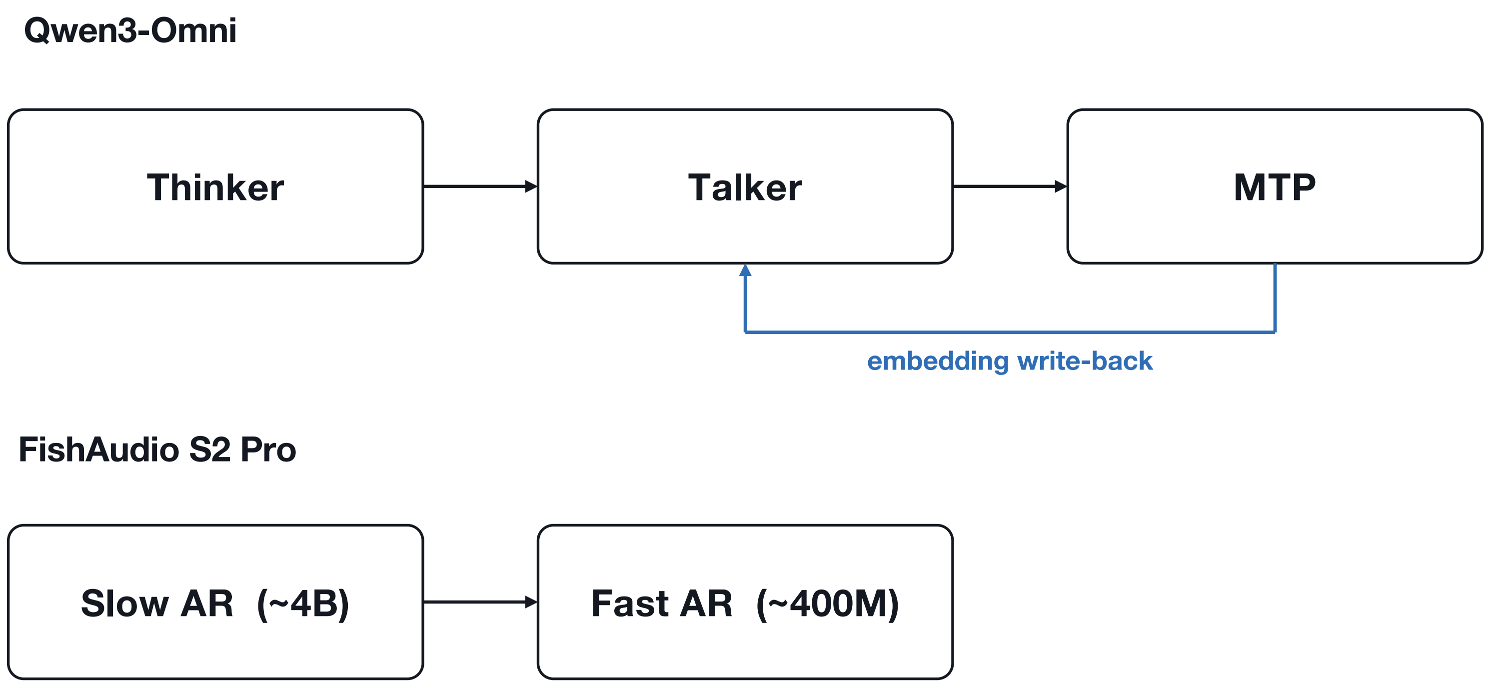
01 What computation process are we optimizing?

02 What computational characteristics does it have?

03 What system did we design?

04 Where are we headed?

Multi-Stage Decoding



Split the Models by the Computing Process

single-stage decoding -> **SGLang**

MiMo Omni, Nemotron Omni
Qwen ASR, MiMo ASR
Wan, Qwen-Image

multi-stage decoding -> SGLang-Omni

Qwen3-Omni, Ming-Omni
TTS
LLaDA-Uni

Follow the first principle of computing, not the modality

Computational Characteristics -> Designs

1 **Scheduler Decoupling**

Heterogeneous bottlenecks -> one Scheduler per stage

2 **Communication Layering**

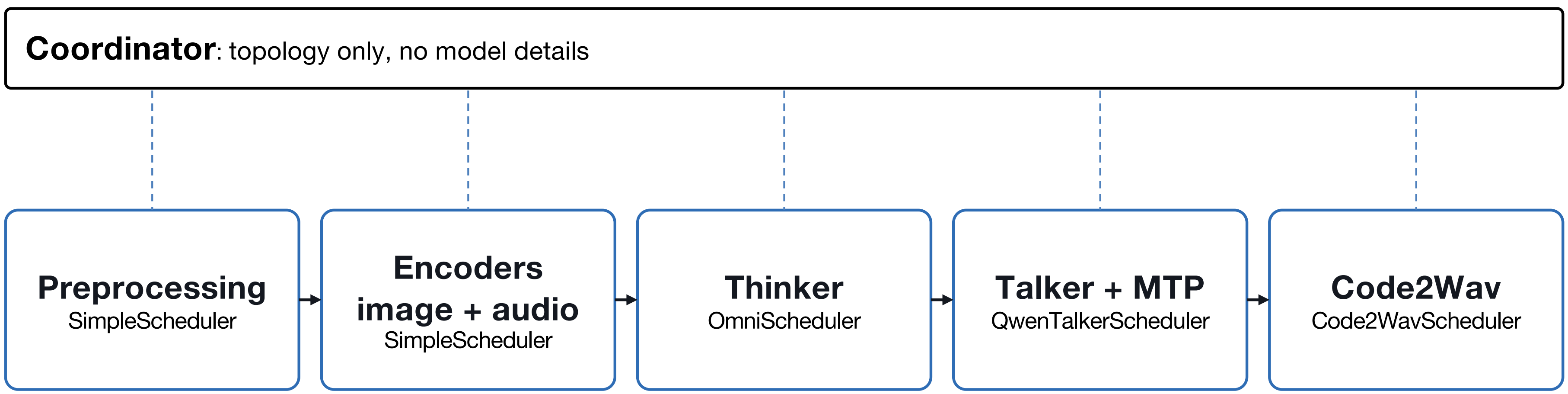
Two kinds of dependency -> control + data plane, tight loop in-stage

3 **Memory Isolation**

Memory contention -> budget + place per stage

1. Scheduler Decoupling

example · Qwen3-Omni



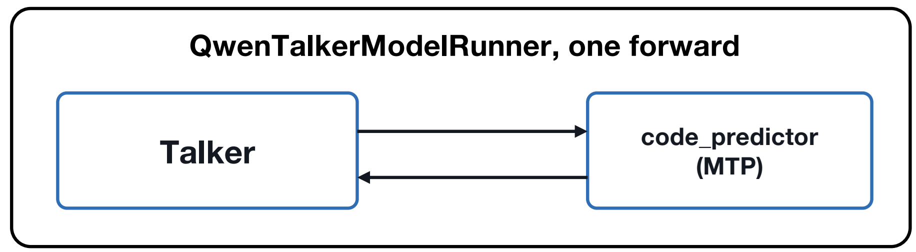
relay = data plane between stages, ZMQ = control plane (Coordinator <-> stage)

2. Communication Layering

split into two Stages

a cross-scheduler round-trip every timestep, latency blows up ❌

one Stage



the Coordinator sees one lightweight decode step

The tight Talker <-> MTP loop never leaves the Stage, no cross-scheduler overhead.

3. Memory Isolation

```
StageConfig(
    name      = "thinker",
    factory   = "...create_thinker",
    wait_for  = ["image_enc", "audio_enc"],
    merge_fn  = "...merge_for_thinker",
    next      = "talker",
    stream_to = ["talker"],
    tp_size   = 2,   gpu = [0, 1],
)
```

Memory is budgeted per stage.

Encoder activation is the hidden cost.
2.5 GB weights, **>30 GB** at peak. TP (tp_size) splits it.

SGLang-Omni Roadmap

- More models
- Shared optimization templates
- Cross-node multi-stage pipeline
- Full diffusion-stage support
- Close the loop with RL
- Production serving

Let’s build together

github.com/sgl-project/sglang-omni

Supported Models

Model	Type
boson-sglang/higgs-audio-v3-tts-4b-base	TTS
fishaudio/s2-pro	TTS
mistralai/Voxtral-4B-TTS-2603	TTS
Qwen/Qwen3-TTS-12Hz-Base	TTS
OpenMOSS-Team/MOSS-TTS-v1.5	TTS
Qwen/Qwen3-Omni-30B-A3B-Instruct	Omni
inclusionAI/Ming-flash-omni-2.0	Omni
inclusionAI/LLaDA2.0-Uni	Multimodal